FGCU QUANTUM HACKATHON · IBM QISKIT FALL FEST 2026

Rewind & Amplify

Grover’s algorithm inside a quantum recurrent state machine



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One click record: 64 steps, 5 clicks

Rare events hold the answers, and cost the most to see

Why it matters

Tiny hidden defects limit today’s qubits, and nobody can put a probe on one. The evidence is in rare patterns of clicks from one qubit’s readout.

Who it hits

The engineers who build and calibrate quantum chips, and anyone paying for quantum time by the second.

The catch

To study what happens around a rare event, you pay for every run in which it didn’t happen.

27.5 million

runs of our model to match one real 64-step click record, once

Median over 4,096 held-out records, from the exact filter

Our model: a quantum recurrent state machine

Drive
The emitter is pushed a little: RY(θ)

Detune
The hidden defect turns: RZ(ζ)

Exchange
They swap energy, by φ

Emit
The emitter may give light, chance γ

Read, then reset
Click or no click: the only thing we see

Repeat every step. Two hidden qubits hold the memory, one bit comes out, and a step costs 4 two-qubit gates on IBM Heron.

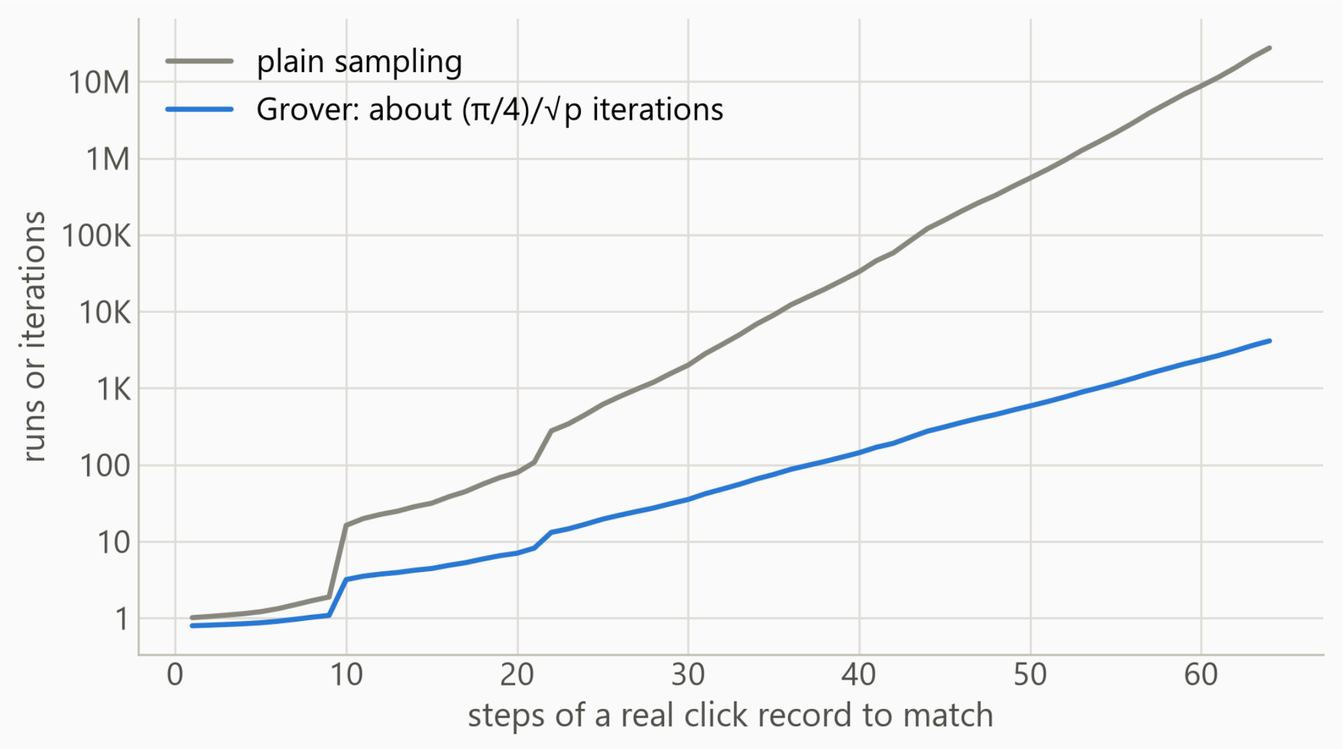
Our work earlier this week

≤ 0.1%
prediction error left when it learns a defect from clicks alone

17%
left by a tuned LSTM on the same records

But
it only generates. You can’t ask it a question, only run it and discard.

Asking it a question means throwing runs away



To condition on a record, you keep only the runs that match it.

The cost is **1/p**, and every step of the record multiplies it.

Grover's algorithm offers **1/√p**. If it can be made to work here.

Median over 4,096 held-out records of the hidden-defect machine, scored by the exact filter

Grover’s algorithm in one minute

1. Start



Every answer is held at once, with equal weight.

2. Mark



Flip the sign of the answers you want.

3. Reflect



Flip everything about the average. The marked answer grows.

Repeat. A **10%** chance becomes **99.9%** in two rounds. Searching N items takes about $\sqrt{\mathbf{N}}$ rounds, where sampling takes N.

$$P = \sin^2((2r + 1)\,\theta)$$
$$p = \sin^2\theta$$

You can’t Grover what you can’t rewind.

A reset can’t be undone

Each Grover round runs the machine backwards. Ours resets a qubit every step.

A state isn’t a database

The hidden memory is compressed. It doesn’t hold records you can address.

Fewer queries isn’t less time

Rewinding and reflecting cost gates, and on real chips gates cost errors.

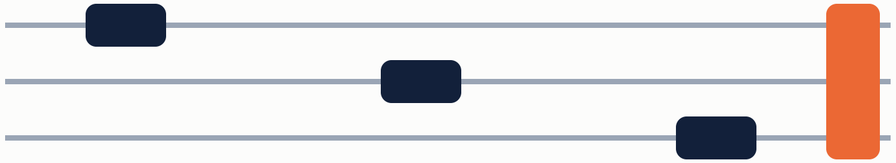
Move 1: Purify. A fresh qubit replaces every reset

Before: one light, reset each step



Measure and reset: cheap in qubits, but the past is gone.

After: a fresh light per step



Nothing measured until the end: one reversible circuit, A.

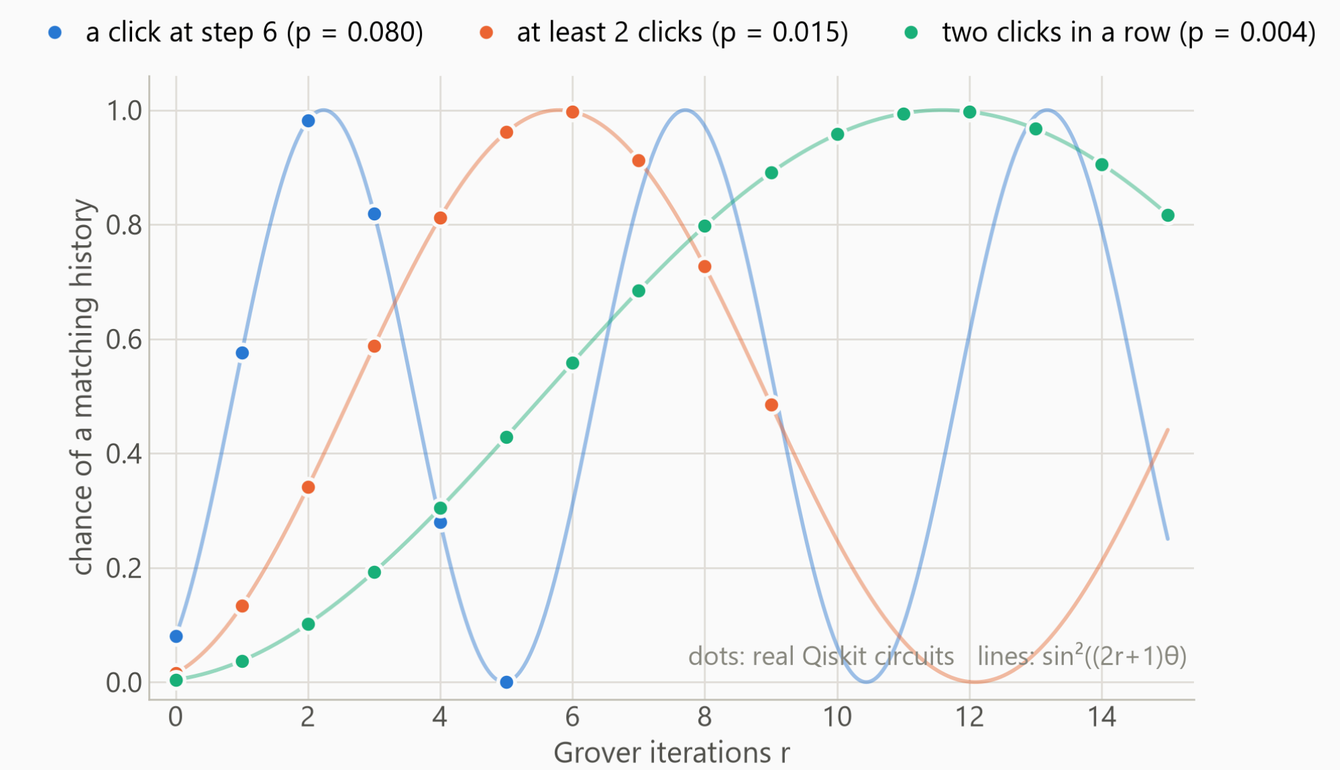
$$A|0\rangle = \Sigma \text{ over } y \text{ of } |m_y\rangle\,|y\rangle$$

The state holds **every click history at once**, each with weight √P(y). Grover's search space is the machine's own past.

4×10^{-16} largest amplitude error against the exact model

p = 0.61 reset and purified circuits click alike, 1,000,000 shots each

Grover now searches the machine’s histories



8×10^{-15}

largest gap from theory, over six questions

Exact

conditioning: matches keep their true relative odds, hidden qubits included

725×

fewer machine calls at $p = 10^{-6}$: slope −0.50 against −1.00

Qiskit grover_operator on the purified QRSM, T = 6, noiseless. Each iteration counts as two machine calls.

Noise punishes depth, so we redesigned around it

456

two-qubit gates: textbook, T = 5

Grover is famously fragile. Each iteration rewinds the whole history and checks every qubit. At 0.2% error per gate, only about 40% of those runs finish clean.

The resilient architecture: four moves

1 Purify

Fresh qubits replace resets, so the machine can be rewound.

2 Steer

Tune one gate so the event is common. Free, and exact.

3 Amplify locally

One iteration, with a reflection on 3 qubits at any length.

4 Verify

The click is measured, so a failed shot shows itself and is dropped.

Move 2: Steer. Part of the rarity is free to remove

A click is rare for two reasons

Did it emit? The emission factor

Set by one gate angle. Turning it up scales every matching history by the same constant, so the answer stays **exact**.

Was it excited? The population factor

No gate setting removes this. It is Grover’s real job.

Tune p to exactly **1/4** and **one** iteration gives certainty, where textbook needs two or three.

Machine calls per conditioned sample of a real 64-step record

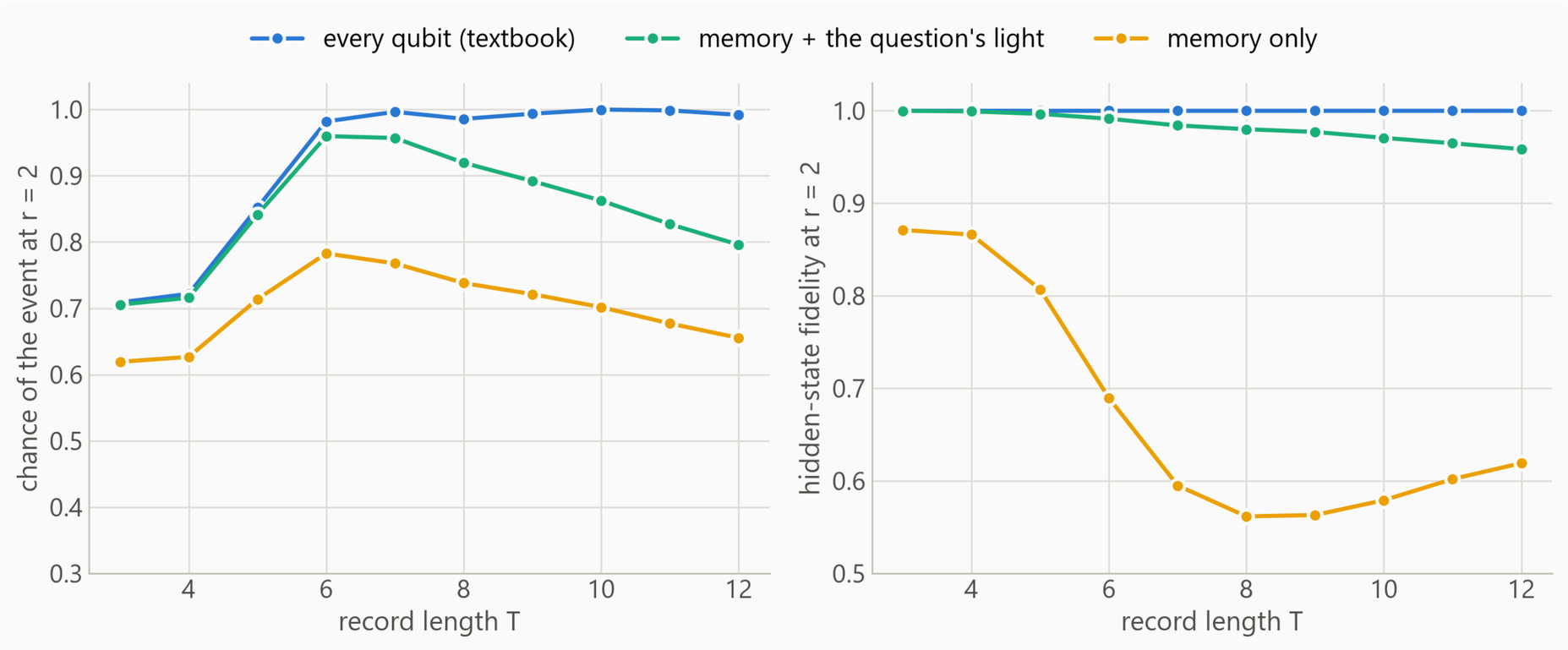
27,495,891
plain sampling

6,240
steered: 4,400× fewer, classical maths

125
steered, then Grover: 50× fewer again

Median of 4,096 held-out records, 5 clicks each. Steering sets every clicked step to emit with certainty. Exact for clicks at known steps.

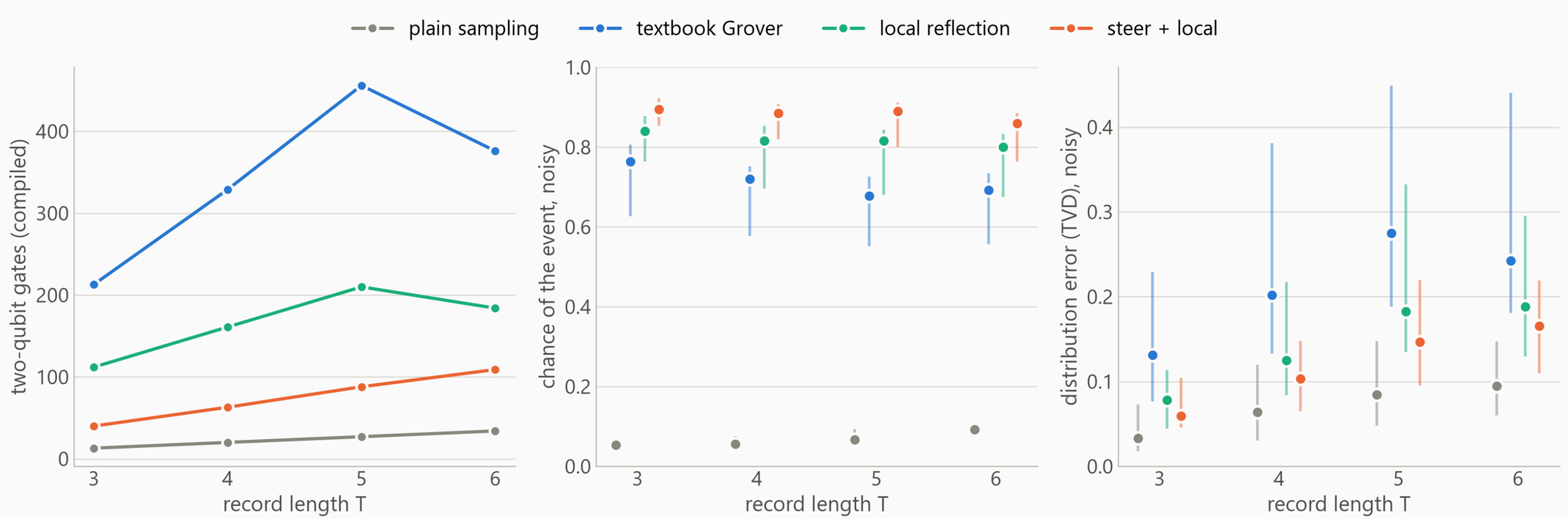
Move 3: Reflect only where the question looks



Watching only the memory looked great on hit rate, then failed: it scrambled the hidden state. Our own test caught it.

Adding the question's light fixes it: **3 qubits** at any length, in place of every qubit. It drifts on long records.

Across five IBM device models: fewer gates, better answers



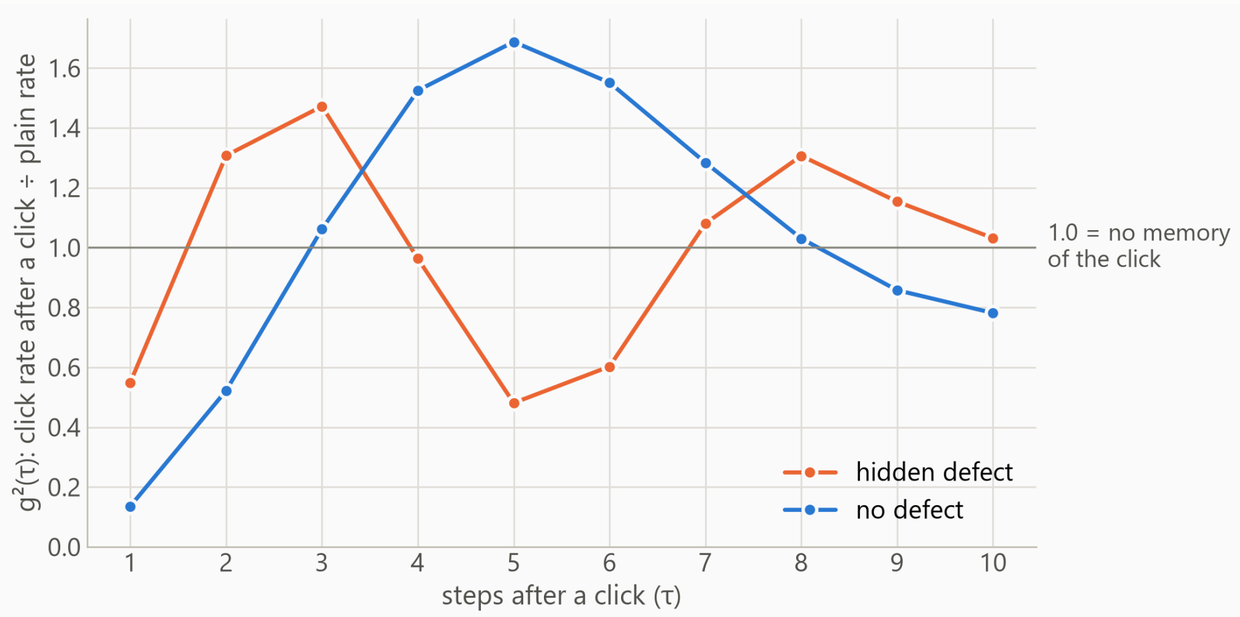
3–5× fewer two-qubit gates than textbook Grover

0.88 chance of the event, against 0.71 for textbook

0.12 distribution error, against 0.21 for textbook

Better on both in all 20 device and length cells. Dots: median; bars: range. Plain sampling keeps only 5–9% of shots.

Now it can answer a physics question



What happens right after a rare click?

With no defect, a click **wipes the machine**: what follows simply restarts the run.

A hidden defect **remembers across the click**. This curve is a fingerprint of hidden memory.

2.4σ → 7.1σ

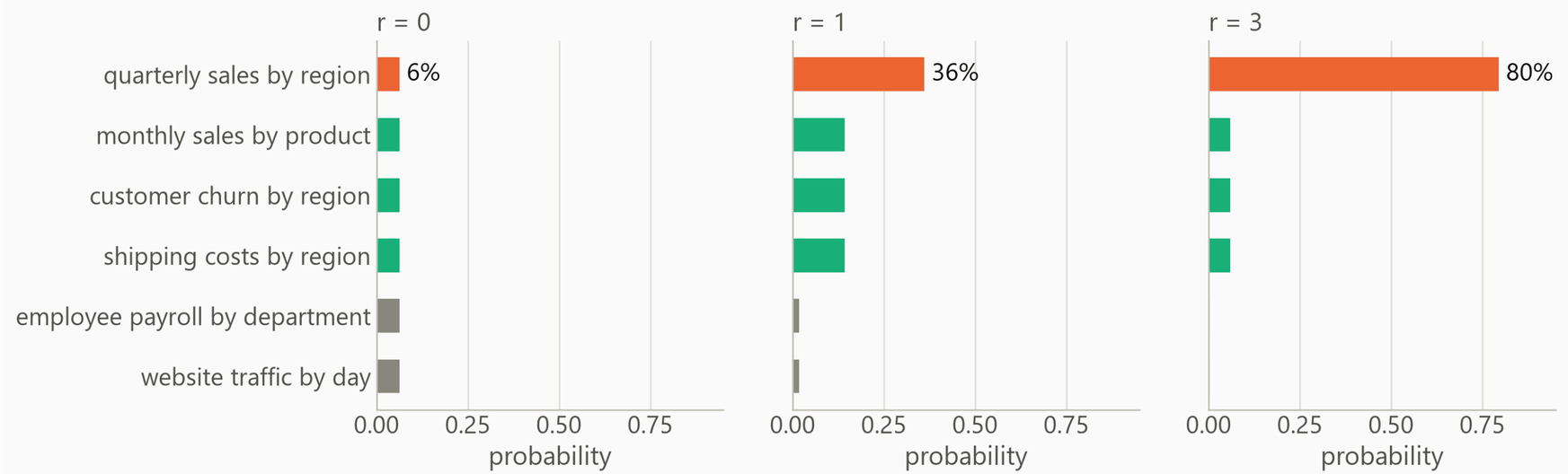
telling defect from no defect at 4,096 shots on noisy device models, once we amplify the click and keep running

Chart: exact curves, click at step 6. Separation: click at step 3, τ = 3, mean of five snapshots, plain sampling against steer + local.

The other slot: a QRSM as the oracle

Grover has two slots: what to search, and what counts as a match. A **reader QRSM** fills the second. It reads a request token by token, turning the phase of each entry that matches.

“how much did we sell in each area”
→ `[sales, region]`



19 of 20 vague requests found the right table; the one miss was the classical tokeniser

No speedup the oracle is compiled from the catalogue: 32,800 gates against a 1,024-step scan

Demo: real circuits, about a minute, no account needed

```
$ python demo/demo.py
A click at step 6: 1,000 shots each
plain sampling                  7.8% usable   distance 0.077
steer + 1 Grover iteration    100.0% usable   distance 0.010
steer + local reflection       96.8% usable   distance 0.054
What follows a click? P(click 1, 2, 3 steps later)
no defect    0.016  0.065  0.096   exact 0.017  0.058  0.101
one defect   0.011  0.027  0.045   exact 0.010  0.030  0.047
“how much did we sell in each area”
quarterly_sales_by_region   81.4%   every entry started at 6.2%
```

Every number comes from a Qiskit circuit run on Aer. `--noisy FakeFez` runs it on an IBM device’s noise model.

The repository also has an interactive lab: watch 48 runs change as you switch builds, and try your own search requests.

Built for real IBM hardware, in proper Qiskit

What is ready

39 circuits in one job, from textbook Grover to the resilient build, with readout calibration and echo noise checks

Two jobs, plain and with dynamical decoupling and Pauli twirling, to measure what suppression buys

Rehearsed end to end on a local device snapshot through the same SamplerV2 code path

Status: not yet run on a real device. Every result in this talk is simulation, including the IBM noise models.

```
qiskit 2.6, aer, ibm-runtime
grover_operator
generate_preset_pass_manager
SamplerV2
dynamical_decoupling, twirling
```

110
automated tests, all passing

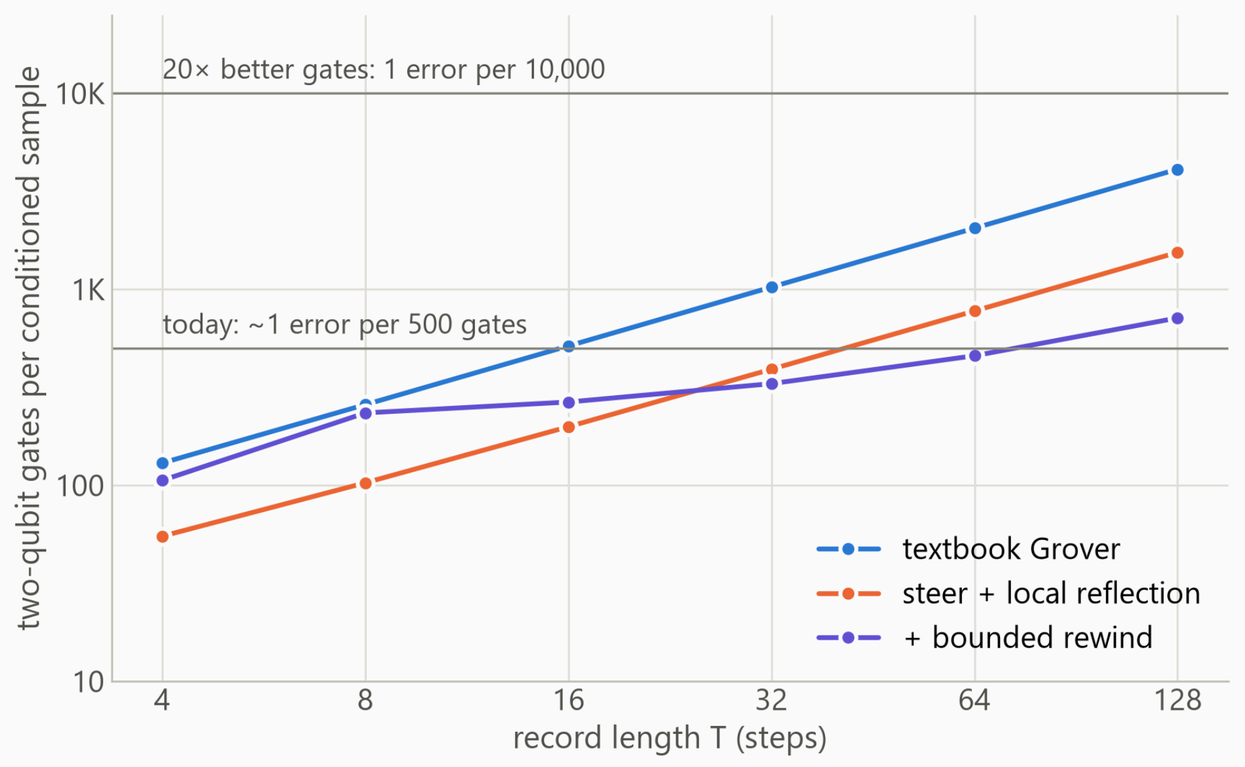
≈ 1 min
estimated QPU time per job

Layouts are chosen by the device’s own error rates: best of 8 compiles at optimization level 3.

What we don’t claim

Claim	The honest position
A speedup	None today. An exact classical sampler gives a sample in about a millisecond here. Its cost grows about 6× per hidden qubit, so a crossover is plausible, but not shown.
Rewinding a real chip	We rewind a model of the chip. A physical defect can’t be run backwards, so this queries what was learned from the clicks.
Steering is quantum	It is classical importance sampling. Exact only for clicks at known steps, and it needs a rough value of p.
Local reflection is exact	It is an approximation. Steered, its hit rate falls to 0.73 at T = 14, where the exact build stays at 1.0.
A new algorithm	Amplified rejection sampling is known (Low, Yoder & Chuang 2014). Ours is the mapping onto hidden quantum models, and the resilience work.

Why it has potential: where the classical baseline runs out



A 66× smaller job

Conditioning on a 64-step record: 2.7 million gates as textbook, 40,000 once steered.

Where classical runs out

IBM's 2029 target, 200 logical qubits and 100 million gates, would fit 50 hidden qubits and 150 steps. That memory alone is about 18 PB classically.

Next

Steering for learned models, error detection, and rewinding only the memory's horizon: 8 steps held hit rate 0.80 to 0.87 out to T = 18.

Scale estimates, not compiled circuits. Chart: logical gates at p = 0.09. Roadmap: IBM's published Starling target.

What we learned, and who did what

What we learned

Reversibility is a design choice. One reset decides whether Grover is possible at all.

Exact is not always best. On noisy hardware a cheaper approximate circuit beat the exact one.

Test what you can’t see. The hit rate looked perfect while the hidden state was scrambled.

The team

Nicholas Taylor: QRSM research, the Grover fusion, experiments

Tim Yudintsev: research, and the keyword search idea

Dan Izquierdo: [add your part]

[Fourth teammate]: [add your part]

Disclosed

The QRSM model predates the event. The Grover fusion is new this week. Built with AI assistance from Claude.

`github.com/[your-account]/qrsm-grover`